COLORADO
Chief Information Officer &
Executive Director
Governor's Office of Information Technology

TS-INF-013: UPS/PDU for Network Systems

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Title	UPS/PDU for Network Systems
Executive Sponsor	Network Services, Network Services Manager
Issuing Authority	Office of Information Technology (OIT)
Applies to	OIT and OIT-supported agencies, employees of OIT, agencies supported by OIT, and vendors contracted by the State of Colorado or its stakeholders.
Tier	Standard
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1. Purpose

This standard is issued under the authority of the Governor's Office of Information Technology (OIT) (C.R.S. 24-37.5, et seq.) and the Office of Enterprise Architecture (Rule 8 CCR 1501-7). It addresses stakeholder needs related to UPS and PDU standards for network communications equipment, providing guidelines for agencies and OIT in selecting the appropriate uninterruptible power supply or power distribution unit based on each agency's specific requirements.

The goal of this standard is to define the minimum requirements for designing, installing, securing, monitoring, maintaining, and protecting UPS systems and PDUs, and to clarify the roles and responsibilities of OIT, vendors, and agencies.

2. Scope

These standards apply to OIT and OIT-supported agencies. This standard shall be used by employees of OIT, agencies supported by OIT, and vendors contracted by the State of Colorado or its stakeholders, to enforce technology standardization and assist in making the usage, support, and/or purchase of technologies more consistent and efficient.

Standards instituted for use by approval of the OIT Architecture Review Board (ARB) govern purchase and deployment decisions deemed "go-forward" or "net-new", and are not intended as a retroactively enforceable mandate for immediate conformance, especially where cost implications may be prohibitive.

3. Roles and Responsibilities

Enterprise Architecture - Chief Enterprise Architect

- Chairs and conducts business of the Architecture Review Board (ARB).
- Updates effective date and/or last review date.
- Publishes the ARB-approved Standard to the OIT public-facing website.

Architecture Review Board (ARB)

- Engages stakeholders and subject matter experts to determine the standard.
- Reviews and approves the Standard (this document).

Network Services - Network Services Manager

- Owner of Standard; engages stakeholders and subject matter experts to determine the standard.
- Updates document standards and version number; presents to the Architecture Review Board for approval and adoption.

All OIT and agency facilities staff are responsible for knowing and following the standards outlined in this document.

4. Standards

This standard for UPS (Uninterruptible Power Supply) and PDU (Power Distribution Unit) outlines the minimum requirements for units that supply power to network equipment. All network communication infrastructure within an agency's OIT support must have power protection, achieved via a protected PDU or a UPS system.

Power Distribution Unit (PDU) Requirements

- Power Protection: PDUs must include built-in surge protection, overload protection, and circuit breakers at each outlet or branch circuit.
- Internal Transfer Switch: The PDU should include an Automatic Transfer Switch (ATS) with minimal delay (preferably under 10 ms) and a manual override option.
- Power Filtering: When a UPS is not used, the PDU must include EMI/RFI filtering capable of filtering frequencies from 150 kHz to 100 MHz.
- Form Factor: Rack-mount PDUs should be designed for standard 19-inch rack mounting (1U or 2U), with modular components.
- Compliance: PDUs should be UL-listed and comply with CE, IEC 60950, and IEC 62040, as well as RoHS regulations.

UPS Topology Options

- Single-Conversion (Standby) UPS: More energy-efficient but less protective; best for smaller setups such as network closets (1–6 switches / 1 router).
- Line-Interactive UPS: Ideal for core locations servicing ISP and backbone fabric for campuses and large buildings.
- Double-Conversion UPS: Offers the highest level of protection; recommended for mission-critical systems.
- Multi-Mode UPS: Balances efficiency and protection; suitable for smaller setups.

UPS Rating

- List all equipment to be protected; determine volts and amps drawn by each device to find maximum load.
- Multiply volts by amps for a VA figure per device; sum all VA figures.
- Multiply the sum by 1.2 to build in room for growth — the recommended UPS rating should equal or exceed this final number.
- Add extra VA capacity for motors, variable-speed drives, or laser printers to account for high power inrush at startup.

Redundant Deployment Architectures

- N+1 Redundancy: One additional UPS unit beyond the number needed to support the load.
- 2N Redundancy: Enough UPS units to support the load in case of a full failure; used where continuous operation is critical.
- 2N+1 Redundancy: Combines N+1 and 2N benefits with extra units beyond the full 2N setup.

5. Variances and Exception Process

Any request to procure or deploy technology that differs from the standards and provisions stated within this document must be requested of, and approved by, the "Owner of Standard" listed under Roles & Responsibilities.

Requests may be acquired by emailing the Director of Network Services, including any required documentation as part of standard procurement or deployment procedures. The request will be evaluated by Network Services leadership, with written approval or denial given within 10 business days, and stored in the Network Services Google Drive.

6. Definitions

UPS (Uninterruptible Power Supply): A device providing emergency power to a load when the primary input power source fails, supplying near-instantaneous protection from batteries.

PDU (Power Distribution Unit): A device with multiple outputs designed to distribute electric power, especially to racks of computers and networking equipment.

Backup Time (Runtime): The amount of time a UPS can supply power to its load from batteries when the main power source is unavailable.

Capacity: The maximum power load a UPS can support, measured in VA or kVA.

Automatic Voltage Regulation (AVR): Stabilizes output voltage without switching to battery power.

Transfer Time: The time it takes for a UPS to switch from mains to battery power; ideally instantaneous or within milliseconds.

SNMP: A protocol for managing devices on IP networks, including UPS systems, enabling remote monitoring and management.